Understanding accessibility

Color & contrast

15.10.2024

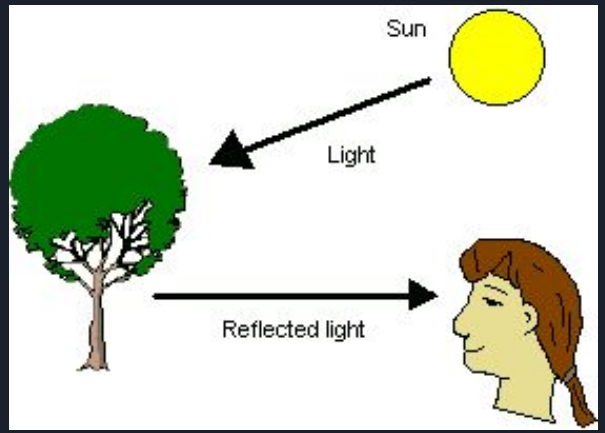
What to expect

- What even is color
- The science of color
- How humans perceive color
- How some humans perceive color differently
- Color contrast, a.k.a. why use of colors matters
- WCAG fun about color and contrast

What is color?

Color - the basics

Color is defined as the aspect of things that is caused by differing qualities of light being reflected or emitted by them. To see color, you have to have light. When light shines on an object some colors bounce off the object and others are absorbed by it. Our eyes only see the colors that are bounced off or reflected. The first person to discover the link between color and light was scientist Sir Isaac Newton.



Color - the basics

Pure white light, such as sunlight, is composed of the visible colors. Sir Isaac Newton discovered this in 1666 by passing a beam of light through a prism.

Although he found an infinite number of colors in this spectrum, Newton wanted to show that there were just seven main colors, like the seven known planets and the seven musical notes in the diatonic scale. He identified red, orange, yellow, green, blue, indigo and violet. This was also in keeping with Aristotle's seven classes of color which he thought were all mixes of black and white.

Using a second prism, Newton showed that each color in the spectrum is monochromatic--that is, composed of a single, unique wavelength which can't be further separated into other colors.

Newton's experiments showed that light can be combined to form different colors. For example, combining blue and yellow light produces a green light that appears identical to the pure green found in a prism spectrum. (Modern techniques, however, show these greens to be two very different colors. Such color pairs are called metamers because they appear to be identical even though they have different wavelengths.)

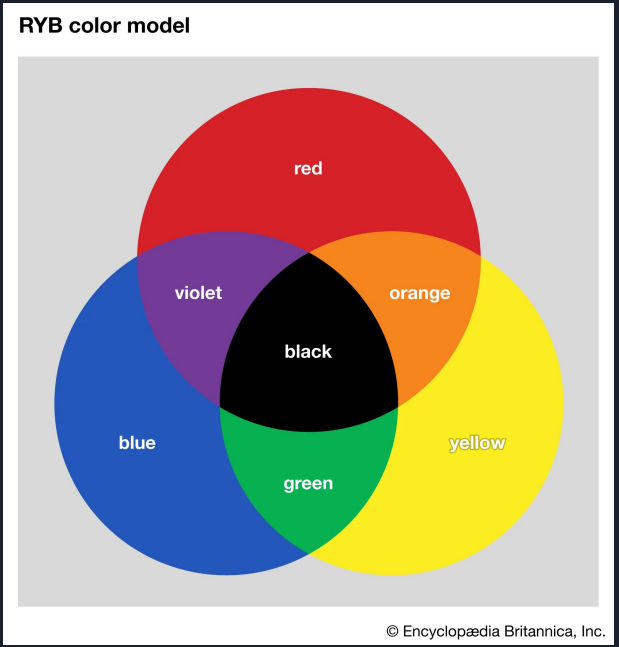
Using two prisms, Newton found that some color combinations produce pure white instead of colored light. In effect, they complete each other when mixed. These pairs of colors are called complements.

Color - the basics

When white light strikes a white object, it appears white to us because it absorbs no color and reflects all color equally. When it strikes a colored object, this color light is reflected back. A black object absorbs all colors equally and reflects none, so it looks black to us. Artists consider black a color, but scientists do not, because black is the absence of all color.

In traditional color theory (used in paint and pigments), primary colors are the 3 pigment colors that cannot be mixed or formed by any combination of other colors. These primary colors are red, yellow and blue. All other colors are derived from these 3 hues.

Secondary colors — orange, green and purple — are formed by mixing the primary colors, and tertiary colors can be produced by mixing a primary and a secondary color.



Color - the basics

When working with color, particularly in design and print, we distinguish two methods: additive and subtractive. On a monitor, display or television screen, colors consist of red, green and blue (RGB). But when it comes to printed media, prints or paint for objects or dyes, cyan, magenta and yellow (CMY) form the basis to compose colors. We call working with RGB the additive use and working with CMY the subtractive use of color.

Additive color – RGB

Additive colors are colors created by mixing two or more colored light sources together. The RGB color system is an example of additive color, as it combines the primary colors, red, green and blue, in various degrees to create a variety of different colors. When all three of the colors are combined to their full extent, they create pure white. However, when they are combined to the lowest degree, they create black.

Subtractive color – CMYK

On the other hand, subtractive colors are created by completely or partially absorbing (also called subtracting) some light wavelengths and reflecting others. Subtractive colors begin as white. As you add filters to the white light, such as ink, this white light takes on the appearance of color. The CMYK color system is an example of subtractive color use, as photos, magazines, and any printed material use subtractive color to reflect and absorb light in order for us to see.

The science of color

The science behind color vision

All light is electromagnetic radiation, but our eyes can only see a small portion of the light spectrum.

For instance, radio waves and gamma rays, two frequencies on opposite ends of the electromagnetic spectrum, are invisible to us.

Visible light is a range of wavelengths between them that humans can actually see. They are called spectral colors, which fall between ultraviolet light and red light, producing the range of colors that we can see.

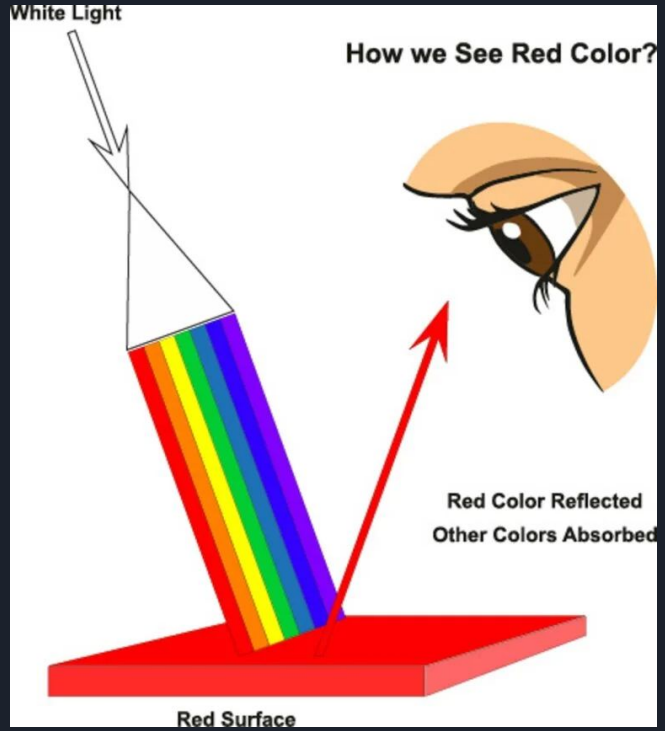
Each color has a different wavelength. Red has the longest (700 nanometers), while violet has the shortest (380 nm) of the visible spectrum.

The science behind color vision

When striking an object, these light wavelengths react differently. Some objects absorb certain wavelengths, while others are reflected (and sometimes, wavelengths are scattered, like in the atmosphere, making the sky blue or red).

For example, if an object appears blue, it's because it absorbed all the other color waves but reflected the blue waves of white light.

If the object is a mixture of colors, it reflects those colors while absorbing the remaining color(s). White objects reflect all the wavelengths, while black objects absorb them—that's why you should avoid wearing dark clothes on hot days.



The science behind color vision

Color comprises three elements, and the combination of these three characteristics is unique for each color.

HUE

The hue of color is red, orange, yellow, green, etc. The hue is also called the “aspect” or “character” of color and is the most discernible attribute in distinguishing different colors.

The hue is often the “richest” version of the color we see.

SATURATION

The saturation (chroma or tone) of a color is how far it deviates from the hue (how much it looks like the “original” color).

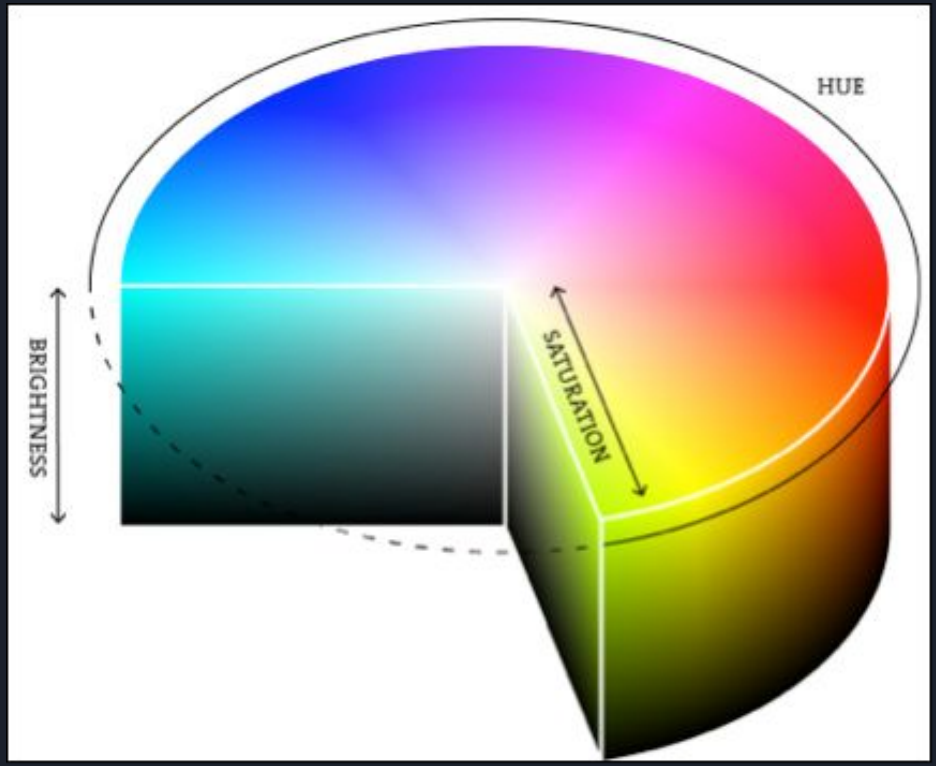
An unmixed bright color is saturated, while a color with white added to it to make it paler or like a pastel color is unsaturated.

Although the color (hue) remains the same, it looks off (faded) from the original, thanks to the saturation.

BRIGHTNESS

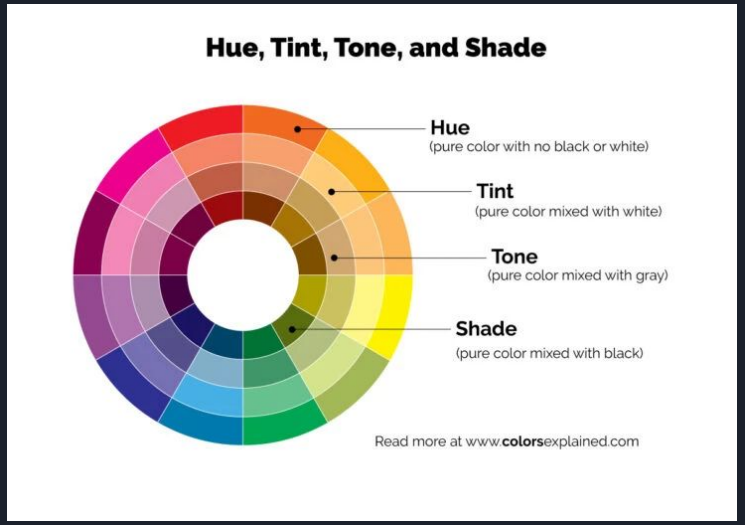
Brightness is the intensity (or value) of the light and how much energy the wave carries. The brightness does not change the hue or saturation level but makes the color appear sharper (more energy) or dull (less energy, like muted colors).

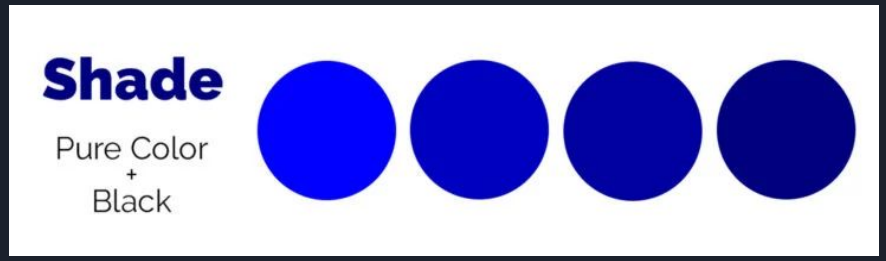
The science behind color vision

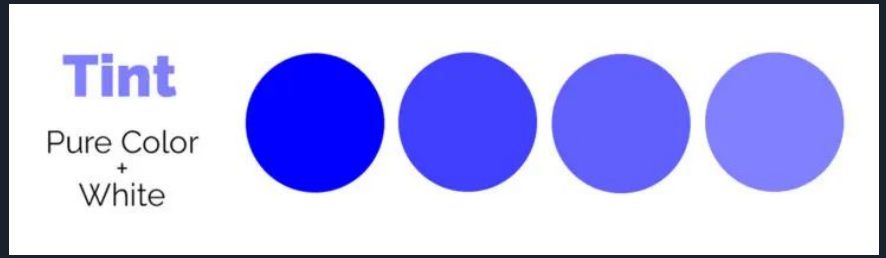


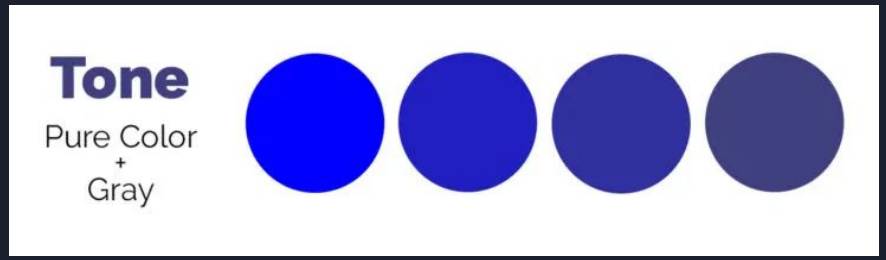
The science behind color vision

A further explanation of how color can be manipulated and how this manipulation is described in each case.







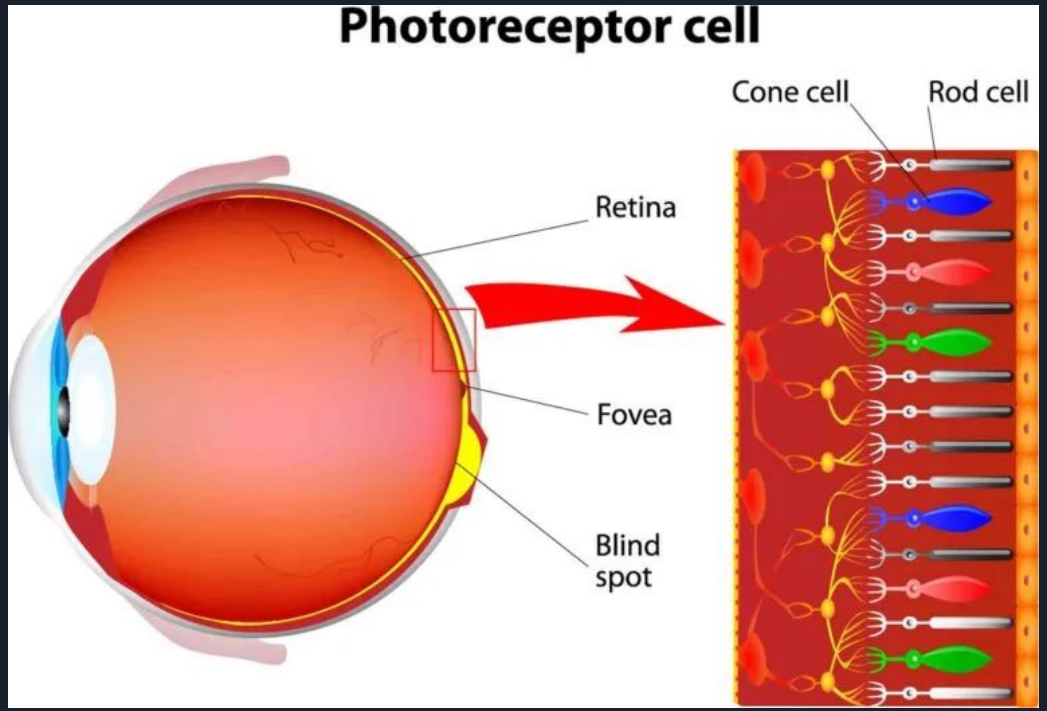


Human visual system

How do we see color?

While color relates to how an object interacts with light, how our brains interpret the information is equally (if not more) important to how we see color.

We recognize color as “color” because of the various structures in our eyes and the processing power of our brains.



How do we see color?

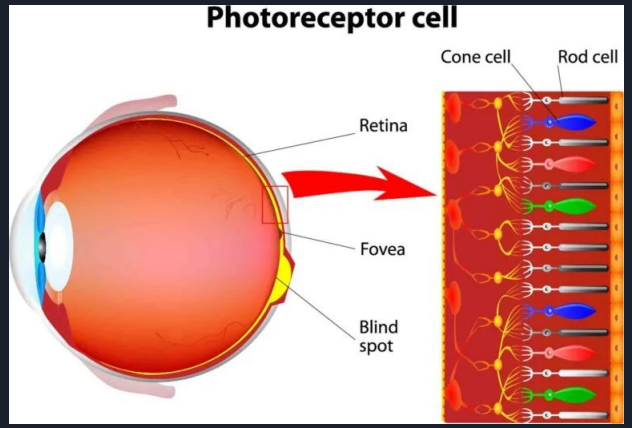
Once visible light reflects off an object, the light enters our eyes through the eye structure, and the light particles (photons) reach the retina at the back of the eye. This critical zone contains photoreceptor cells for processing visual stimuli (information).

The two prominent types of photoreceptors are rods and cones.

Although humans have approximately 100 million rod cells, they are not involved with color differentiation. Rod cells help us see in dim light situations (night vision). Rods require low levels of photons to activate (scotopic vision).

There are roughly 6 million cone cells in the retina, divided into red, green, and blue cone types (pigments). For further image clarity and sharpness, many of these cone cells are clumped together in the “fovea” (a small hollow in the central portion of the retina).

Cones require more energy to activate (photopic vision).



How do we see color?

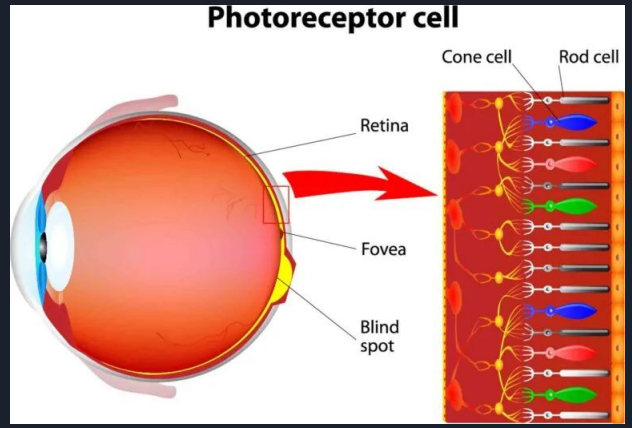
These receptors “receive” photons that activate (supply energy to) proteins in the cell. Once activated, these proteins pass chemical and electrical signals (impulses) to the optical nerve, which transports the information to the brain.

A cone cell contains the protein “photopsins,” while rods contain “rhodopsin.”

Further out, there are three types of cones (pigments) that specialize in recognizing/processing red (long wavelength sensitive cones), green (middle wavelength sensitive), and blue (short wavelength sensitive).

Each of these cones recognizes roughly 100 different shades. However, there is a degree of overlap between the cones, which allows the eye to recognize other colors. Researchers believe the human eye can see over 1 million different colors.

These photoreceptors are “back-to-front,” so light energy moves to the eye’s posterior, where the receptors absorb the light. This strange orientation allows the surrounding cells to replenish the proteins used in communication and protects the eye from intense light.



How do we see color?

Seeing color relates to more than the work of the eye. For example, when white light illuminates a red apple, the apple absorbs green and blue light while reflecting the red.

This reflected wavelength finds its way to our eyes. The light enters and activates the red-receptor cones (thanks to the longer wavelength), which become energized and release the photopsin proteins. These proteins move through the optic nerve to the brain's occipital lobe.

In the brain, the nerve impulse reaches the visual cortex (after passing through the thalamus). A different hemisphere of the brain processes each eye's information.

Although the process is not fully understood, the visual cortex splits the information. It passes it to other regions in the brain, including the prefrontal cortex, which deals with memories, emotions, and sensory information to understand what you're looking at.

Color relates to memory, so even though the lighting might change, our brains recognize and "remember" how a certain color looks and fill in the gaps. Our brains recognize the red apple as red thanks to the stimuli and the brain's processing power.

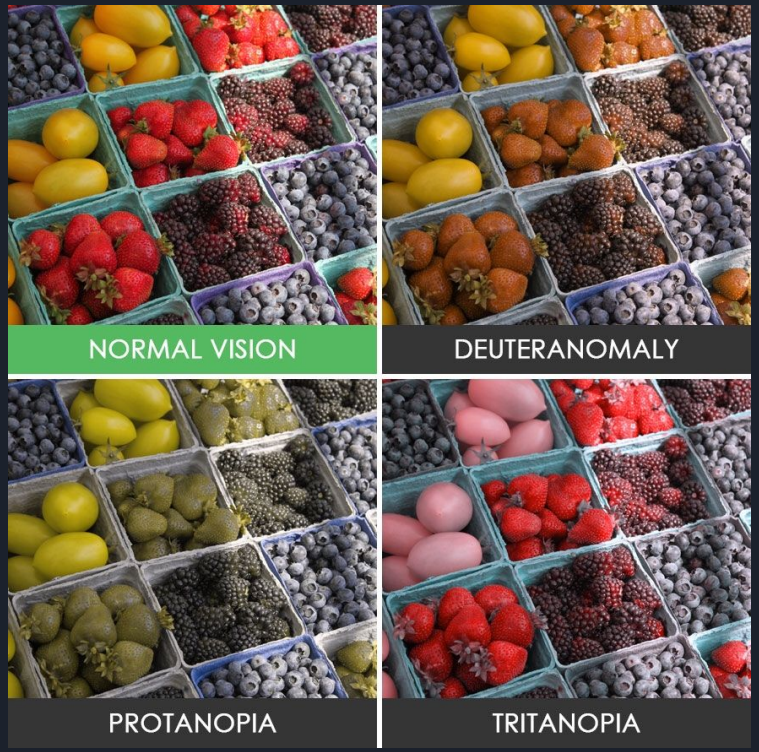
Color is subjective

Perception differences and color accessibility

Color perception refers to how a color subjectively appears or looks to the beholder.

Not all eyes and brains are the same and as a result we may perceive color differently. Still, how do we see color differently, and why?

People interpret the same color differently for various reasons.



Perception differences and color accessibility

According to a study in 2010, women see more shades than men, particularly in the red and green colors. Women are also less likely to suffer from color impairments, like color blindness.

Color blindness affects roughly 1 in 200 women, while every 1 in 12 men is affected.

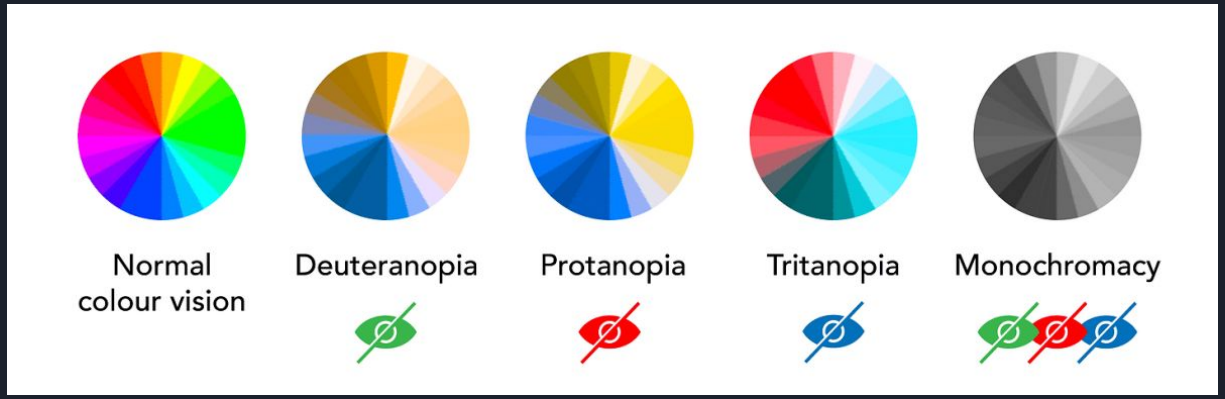
Color vision deficiency (color blindness) is a condition where individuals struggle to differentiate between certain shades/hues.

This blindness is due to fewer certain color-sensitive pigments in the cones (i.e., fewer red, green, or blue-sensitive pigments). This lack of pigment means the eye is not as efficient at differentiating certain wavelengths of light.

The most common form is red-green blindness, while blue-yellow is rare. In exceptional cases, someone might have monochromacy (complete color blindness).

Perception differences and color accessibility

Color blindness is most often a genetic condition; however, other circumstances can prevent people from seeing color, including trauma (which makes it possible to be color blind in one eye and have normal color vision in the other eye), eye and neurological conditions (like glaucoma and multiple sclerosis), side effects of certain medications or exposure to harmful chemicals, diabetes or cancer

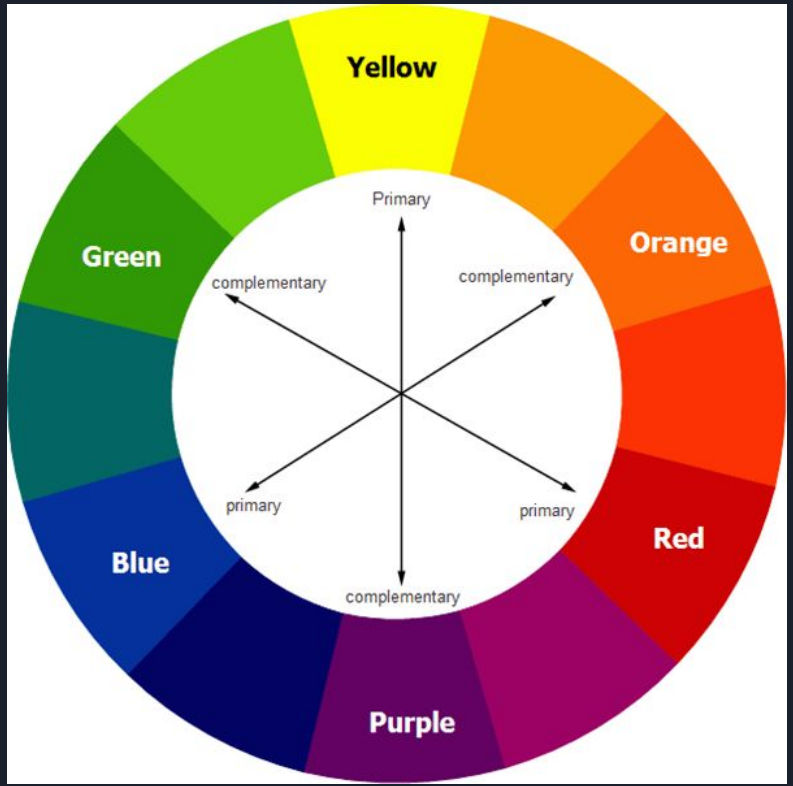


Color contrast

Color contrast

Two colors from different segments of the color wheel are contrasting colors (also known as complementary or clashing colors). For example, red is from the warm half of the color wheel and blue is from the cool half. They are contrasting colors.

In science and color theory, there are precise definitions for contrasting and complementary colors and how they appear on the color wheel. In graphic design and some other fields, we use a looser interpretation. Colors don't have to be direct opposites or have a set amount of separation to be considered contrasting or complementary. In design, it's more about perception and feeling.



Color contrast

Color contrast refers to the difference in luminance between text and background elements.

LUMINANCE

Luminance is a photometric measure of the luminous intensity per unit area of light travelling in a given direction. It describes the amount of light that passes through, is emitted from, or is reflected from a particular area, and falls within a given solid angle.

BRIGHTNESS

Brightness is the intensity (or value) of the light and how much energy the wave carries. The brightness does not change the hue or saturation level but makes the color appear sharper (more energy) or dull (less energy, like muted colors).

Brightness is the term for the *subjective* impression of the *objective* luminance measurement standard.

In other words, brightness is your personal perception of luminance coming directly from a light source or reflecting from something.

Color contrast

So, as said, color contrast refers to the difference in luminance between text and background elements.

In our line of work, luminance is emitted from the screens that we use. How that luminance is understood depends both on the components displayed, a.k.a. the user interface, as well as the receiving “components”, a.k.a. our eyes.

Good color contrast is the basis of good UI design and a critical part of accessibility.

Color contrast in practice

WCAG criterias explained

So how do you actually work with contrast and how it is measured? The answer might not surprise you.

WCAG has a few different criteria relating to both color and contrast. Failing to meet these criteria will mark your design as failure in the eyes of accessibility. The thing is, passing them doesn't guarantee accessibility either, it simply means that this one part of it is handled. While contrast is important, so are plenty of other things.

WCAG has three different acceptance criteria:

- A - you absolutely need to have this, this is the barest form of accessibility
- AA - the level you should always aim for, the "standard"
- AAA - the top level of accessibility that not a lot of software reach

WCAG criterias explained

For color and contrast, these are the criteria you need to work with (WCAG 2.2)

A

- 1.4.1 Use of Color

AA

- 1.4.3 Contrast (Minimum)
- 1.4.11 Non-text Contrast

AAA

- 1.4.6 Contrast (Enhanced)
- 1.4.8 Visual Presentation
- 2.4.13 Focus Appearance

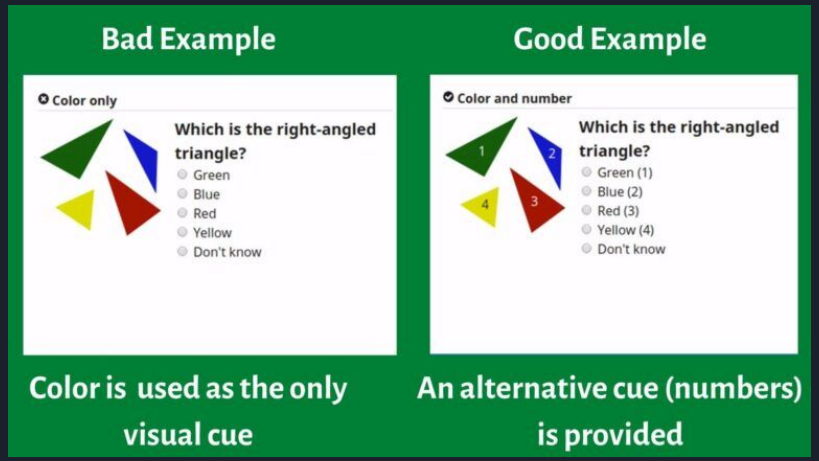
WCAG criterias explained

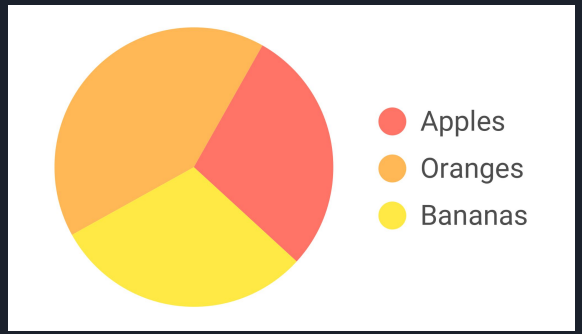
Level A - 1.4.1 Use of color

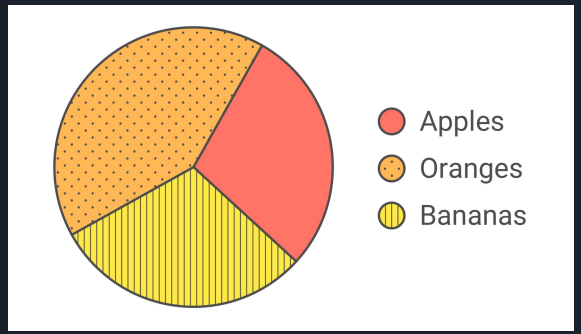
Goal
Color is not the only way of distinguishing information.

What to do
Use information in addition to color, such as shape or text, to convey meaning.

Why it's important
Not everyone sees colors or sees them the same way.







WCAG criterias explained

Level AA - 1.4.3 Contrast (minimum)

Goal

Text can be seen by more people.

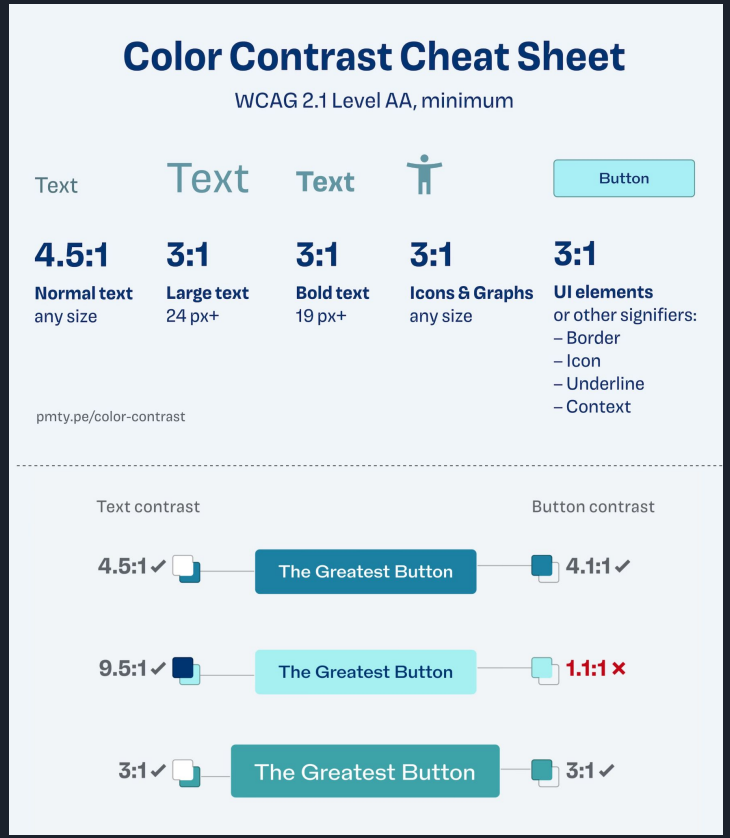
What to do

Provide sufficient contrast between text and its background.

Why it's important

Some people cannot read faint text.

NOTE: User Interface Components that are not available for user interaction (e.g., a disabled control in HTML) are NOT REQUIRED to meet contrast requirements. An inactive user interface component is visible but not currently operable. An example would be a submit button at the bottom of a form that is visible but cannot be activated until all the required fields in the form are completed.



WCAG criterias explained

Level AA - 1.4.11 Non-text contrast

Goal
Important visual information meets the same minimum contrast required for larger text.

What to do
Ensure meaningful visual cues achieve 3:1 against the background.

Why it's important
Some people cannot see elements with low contrast.

Success Criterion (SC)

The visual presentation of the following have a contrast ratio of at least 3:1 against adjacent color(s):

User Interface Components

Visual information required to identify user interface components and states, except for inactive components or where the appearance of the component is determined by the user agent and not modified by the author;

Graphical Objects

Parts of graphics required to understand the content, except when a particular presentation of graphics is essential to the information being conveyed.

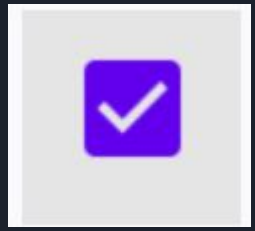
WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

Button Button

Figure 1

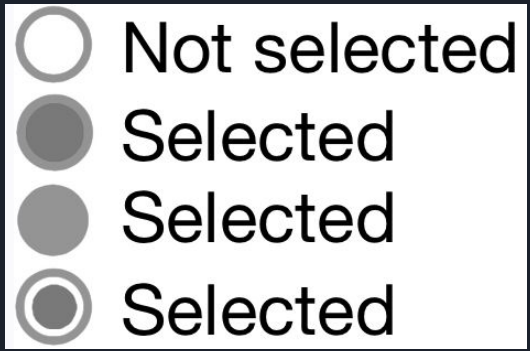
Pass: A button without a visual boundary, and the same button with a focus indicator that is a defined visual boundary of grey (#949494) adjacent to white.



Name:

Figure 3

Pass: The contrast of the input background (white) and color adjacent to the control (dark blue #003366) is sufficient. There is also a border (silver) on the component that is not required to contrast with either.



WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

Figure 6

Pass: Two examples which pass this success criterion, using either a solid fill to indicate a checked-state that has contrast, or a thicker border as well as yellow fill.

Figure 7

Fail: The first example fails the Use of color criterion due to relying on yellow and white hues. The second example fails the Non-text contrast criterion due to the yellow (`#FFF000`) to white contrast ratio of 1.2:1.

WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

Figure 8

Pass: The internal yellow indicator (#FFFF00) contrasts with the blue button background (#4189B9).

Figure 9

Fail: The external yellow indicator (#FFFF00) does not contrast with the white background (#FFF) which the component is on.

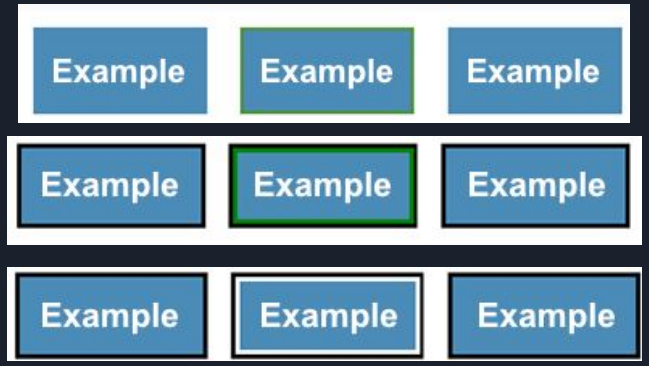
Figure 10

Pass: The external green indicator (#008000) does contrast with the white background (#FFF) which the component is on. It does not need to contrast with both the component background and the component, as visually the effect is that the button is noticeably larger, and it's not necessary for a user to be able to discern this extra border in isolation. Although this passes non-text contrast, it is not a good indicator unless it is very thick. New in WCAG 2.2: There is a AAA criterion in WCAG 2.2 that addresses this aspect, Focus Appearance.

If an indicator is partly inside and partly outside the component, either part of the indicator could provide contrast.

Figure 11

Pass: The focus indicator is partially inside, partially outside the button. The internal part of the yellow indicator (#FFFF00) contrasts with the blue button background (#4189B9).



WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

HOVER STATES

The language of Non-text Contrast specifically calls out "visual information required to identify...states." When users talk about a hover state, they are normally referring to a visual effect that takes place when the pointer is positioned over a control. However, there are a number of HTML components (such as buttons, checkboxes, radio buttons, and selects) which do not by default display any additional visual effects when the user moves a pointer control over them. The pointer itself, via its location, is the indicator of whether the user is hovering on a component. Therefore, additional author-supplied visual treatments for hover are not "required to identify" the hover state. Those treatments can be considered supplemental and do not themselves need to contrast 3:1 against the background.

WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

GRAPHICAL OBJECTS

The term "graphical object" applies to stand-alone icons such as a print icon (with no text), and the important parts of a more complex diagram such as each line in a graph. For simple graphics such as single-color icons the entire image is a graphical object. Images made up of multiple lines, colors and shapes will be made of multiple graphical objects, some of which are required for understanding.

Not every graphical object needs to contrast with its surroundings - only those that are required for a user to understand what the graphic is conveying. Gestalt principles such as the "law of continuity" can be used to ignore minor overlaps with other graphical objects or colors.

WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

“REQUIRED FOR UNDERSTANDING”

The term "required for understanding" is used in the Success Criterion as many graphics do not need to meet the contrast requirements. If a person needs to perceive a graphic, or part of a graphic (a graphical object) in order to understand the content it should have sufficient contrast. However, that is not a requirement when:

- A graphic with text embedded or overlayed conveys the same information, such as labels *and* values on a chart. Note: Text within a graphic must meet 1.4.3 Contrast (Minimum).
- The graphic is for aesthetic purposes that does not require the user to see or understand it to understand the content or use the functionality.
- The information is available in another form, such as in a table that follows the graph, which becomes visible when a "Long Description" button is pressed.
- The graphic is part of a logo or brand name (which is considered "essential" to its presentation).

WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

Gradients

Gradients can reduce the apparent contrast between areas, and make it more difficult to test. The general principles is to identify the graphical object(s) required for understanding, and take the central color of that area. If you remove the adjacent color which does not have sufficient contrast, can you still identify and understand the graphical object?

Figure 17

Removing the background which does not have sufficient contrast highlights that the graphical object (the "i") is not then understandable.

WCAG criterias explained

Level AA - 1.4.11 Non-text contrast continued

ESSENTIAL EXCEPTIONS

Graphical objects do not have to meet the contrast requirements when "a particular presentation of graphics is essential to the information being conveyed". The essential exception is intended to apply when there is no way of presenting the graphic with sufficient contrast without undermining the meaning. For example:

- Logotypes and flags: The brand logo of an organization or product is the representation of that organization and therefore exempt. Flags may not be identifiable if the colors are changed to have sufficient contrast.
- Sensory: There is no requirement to change pictures of real life scenes such as photos of people or scenery.
- Representing other things: If you cannot represent the graphic in any other way, it is essential. Examples include:
    - Screenshots to demonstrate how a website appeared.
    - Diagrams of medical information that use the colors found in biology (example medical schematic from Wikipedia).
    - color gradients that represent a measurement, such as heat maps (example heatmap from Wikipedia).

Tools for color contrast measurement

Tools for measuring color contrast:

- WebAIM contrast checker: https://webaim.org/resources/contrastchecker/
- WAVE evaluation tool: https://chromewebstore.google.com/detail/wave-evaluation-tool/jbbplnpkjmmeebjpijfedlgcdilocofh
- WCAG color contrast checker: https://chromewebstore.google.com/detail/wcag-color-contrast-check/plnahcmalebffmaghcpcmpaciebdhgdf
- Figma plugin Stark: https://www.figma.com/community/plugin/732603254453395948/stark-contrast-accessibility-checker

When in doubt, WCAG 2.2: https://www.w3.org/TR/WCAG22/

All done, thank you!